DESIGN AND IMPLEMENTATION OF AN INTELLIGENT INTERVIEW SIMULATION SYSTEM (IISS)

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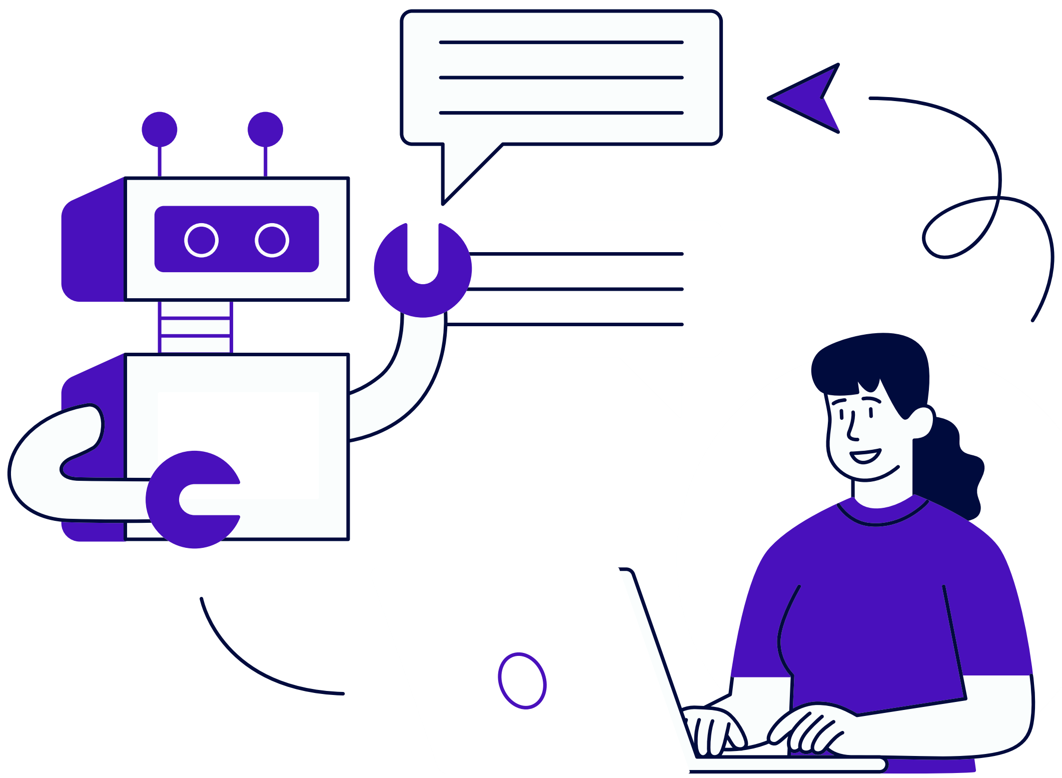

Meet Our Team

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Agenda

- Introduction
- Problem Definition
- Objectives
- Previous Work
- Proposed System
- System Implementation
- System Demonstration
- Conclusions & Future Work

Introduction

Introduction

- Normal interviews play a vital role in evaluating a candidate’s communication skills, personality, and overall readiness for a job

- With the rise of advanced technology, artificial intelligence has become capable of understanding speech, analyzing behavior, and interacting with users in real time

- When combining both, AI-powered systems can now simulate realistic interview experiences by evaluating candidates’ behavioral responses, communication abilities, and knowledge levels while providing intelligent feedback to help users improve their overall interview performance.

Problem Definition

Problem definition

- **Candidates Lack Realistic Interview Practice Area**

Candidates do not have an effective way to practice real interview scenarios.

- **Limited Feedback for Candidates**

Candidates rarely receive detailed feedback after interviews.

- **Time-Consuming for Recruiters**

Recruiter struggles to efficiently manage and assess a high number of applicants.

- **Difficulty in Tracking Candidates Progress**

Candidates struggle to identify areas needing improvement across different interview attempts.

- **Security and Integrity Challenges in the Hiring Process**

Recruiters face difficulties in preventing cheating, identity manipulation, and unauthorized assistance during online interviews and assessments, which can affect the fairness and reliability of candidate evaluations.

Objectives

Objectives

Realistic Interview Simulation (Visual & Audio Interaction)

Dynamic Real-Time Conversation

Intelligent Mock Interviewer

Behavioral & Emotional Performance Analysis

Objectives

Intelligent Resume Context Extraction

Adaptive Question Source Configuration

Comprehensive Feedback & Evaluation Reports

Advanced Anti-Cheating & Identity Verification

Previous Work

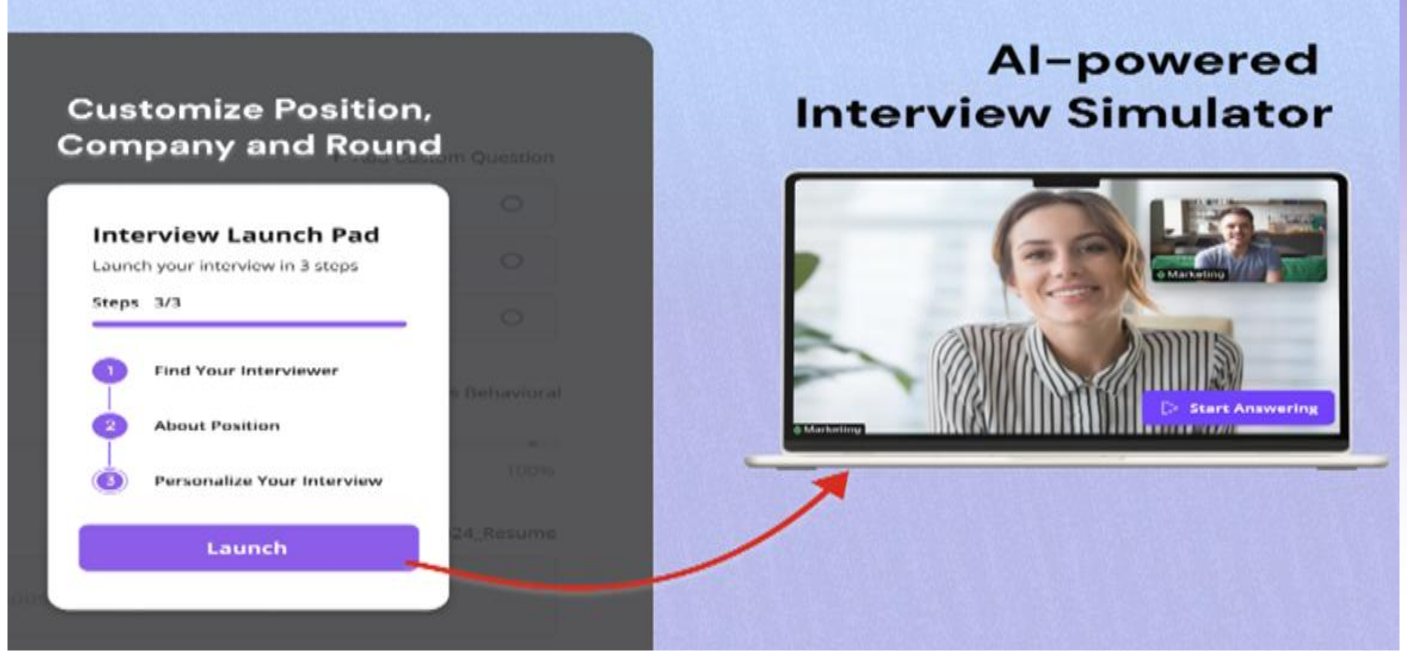
Previous Work

- Overview of existing interview simulation and AI evaluation systems.
- Review of platforms across web, desktop, and mobile environments.
- Summary of core methods, tools, and techniques used in past systems.
- Functional analysis of system operation, strengths, and limitations.
- Comparison foundation: identifying gaps and opportunities.
- Context for the need of enhanced AI detection, classification, and adaptive simulation features.

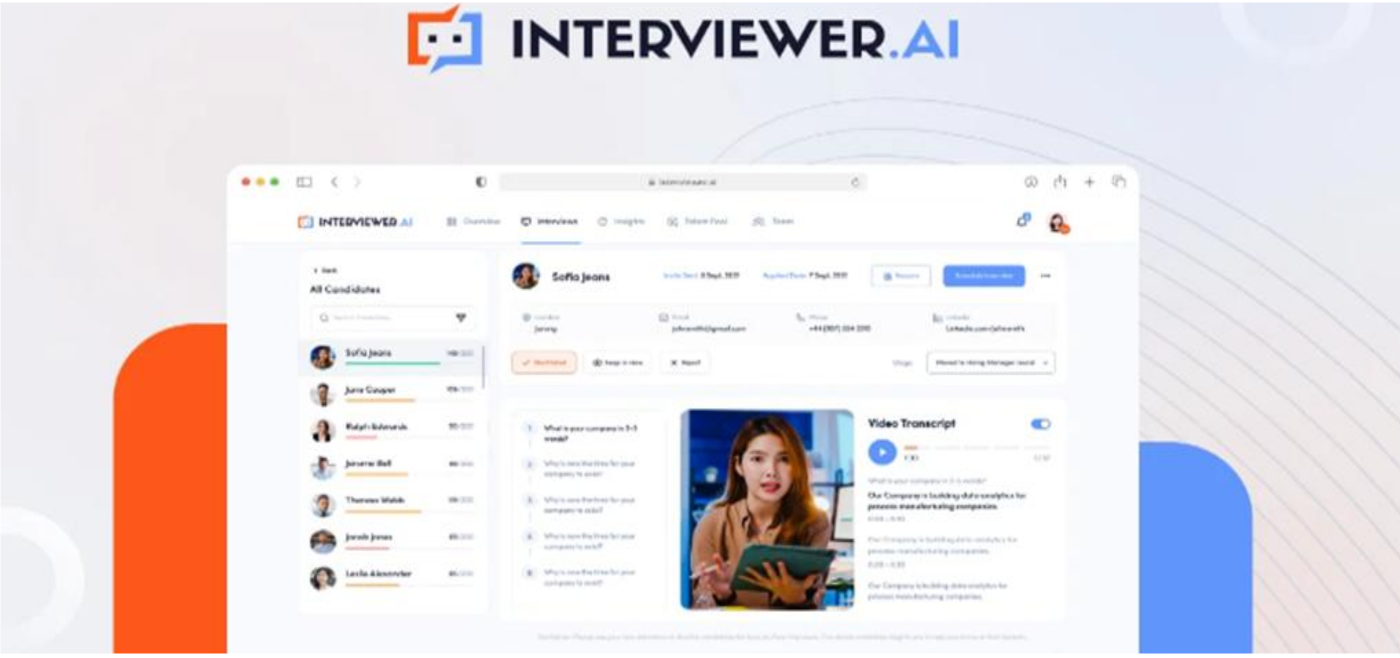
Previous Work (Cont.)

Websites

- AMA Interview AI **[1]**
  An AI-powered website that helps users practice job interviews through real-time conversation simulations and instant performance feedback.



- Interviewer AI **[2]**
  An advanced website designed for automated candidate assessment through AI-driven interviews, voice analysis, and personality insights.

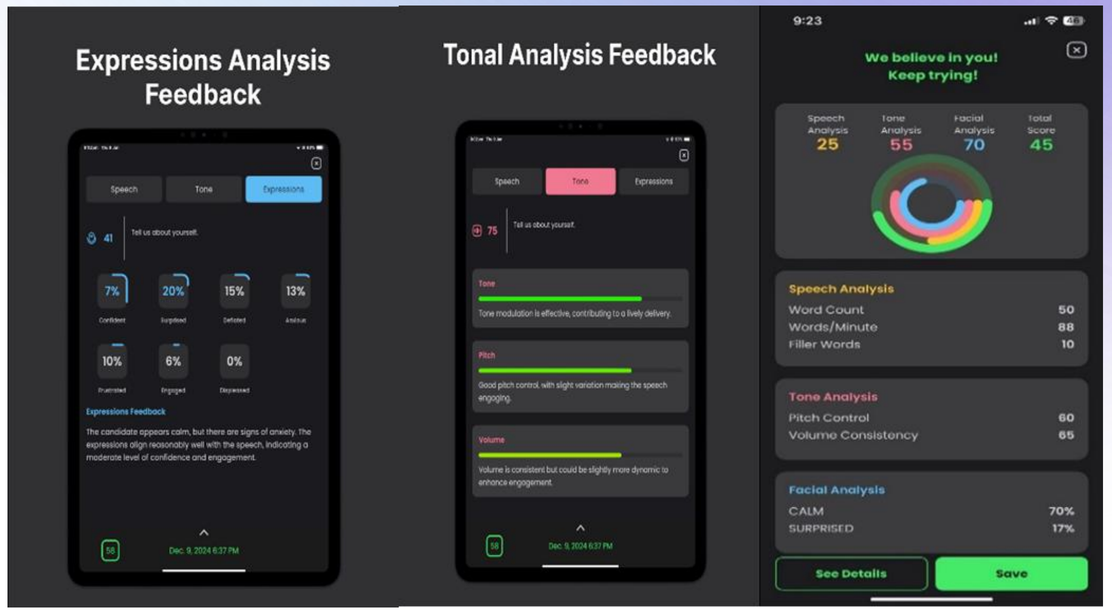


Previous Work (Cont.)

Mobile Applications

- Konfidence **[3]**

 AI Interview Coach A mobile application that helps users build confidence through AI-based interview practice, personalized tips, and video analysis.



- Mocktok **[4]**

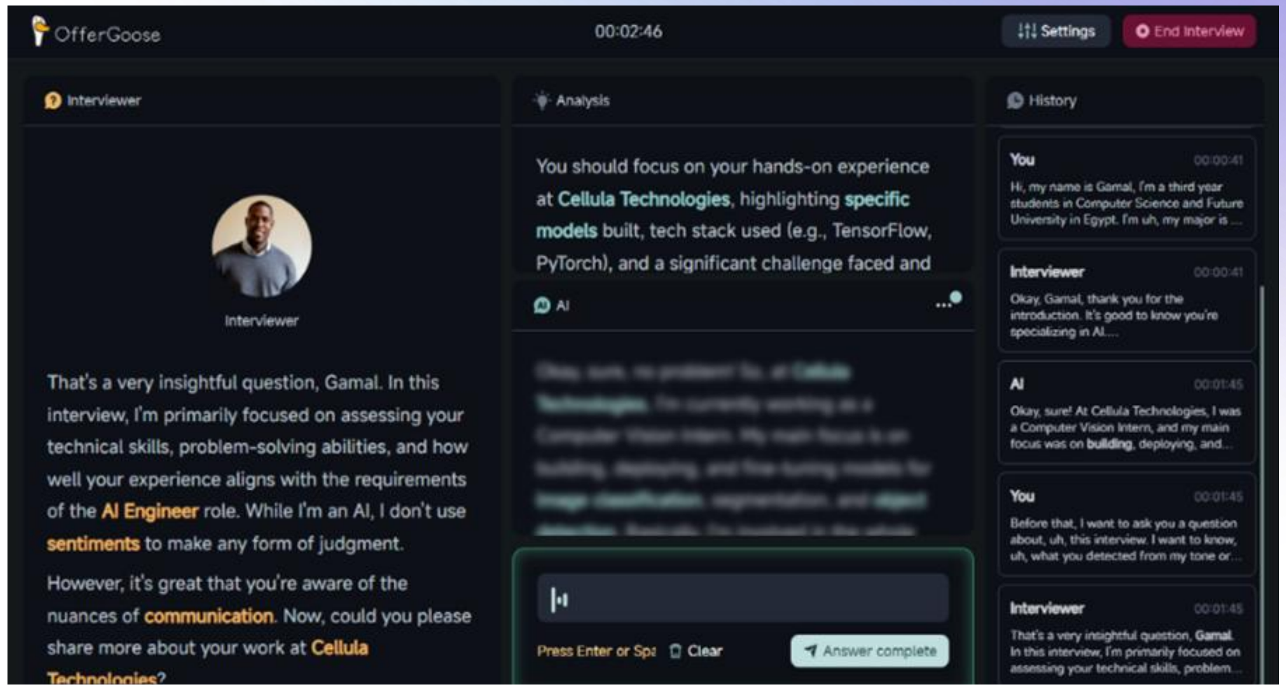
A mobile application that allows users to simulate interviews with AI, track their performance, and receive detailed feedback.

Previous Work (Cont.)

Desktop Applications

- OfferGoose **[5]**

 A desktop application that uses AI to optimize resumes, prepare users for job interviews, and improve their overall hiring performance.



- InterviewAgent AI **[6]**

 A desktop application offering AI-driven mock interviews, recruiter-style evaluations, and automated improvement reports.

Comparison (Mobile)

Features \ Apps	Konfidence AI	Mocktok	Proposed System
Facial expression detection	☑	✗	☑
Tone expression detection	☑	✗	☑
Generating feedback	☑	☑	☑
Video Accessibility	☑	☑	☑
Audio answering	☑	☑	☑
Real Conversation	✗	✗	☑
Mock Avatar	✗	✗	☑
Storage of user data	☑	☑	☑
Specifying Tracks	✗	✗	☑
Extract user data from resume	✗	✗	☑
Multi-Face Detection	✗	✗	☑
Mobile Detection	✗	✗	☑
Identity Verification	✗	✗	☑

Comparison (Web)

Features \ Apps	AMA	Interviewer.AI	Proposed System
Facial expression detection	✗	☑	☑
Tone expression detection	✗	✗	☑
Generating feedback	☑	☑	☑
Video Accessibility	☑	☑	☑
Audio answering	☑	☑	☑
Real Conversation	✗	✗	☑
Mock Avatar	☑	☑	☑
Storage of user data	☑	☑	☑
Specifying Tracks	✗	✗	☑
Extract user data from resume	✗	☑	☑
Multi-Face Detection	✗	✗	☑
Mobile Detection	✗	✗	☑
Identity Verification	✗	✗	☑

Comparison (Desktop)

Features \ Apps	OfferGoose	Interviewer Agent	Proposed System
Facial expression detection	✗	✗	☑
Tone expression detection	✗	☑	☑
Generating feedback	☑	☑	☑
Video Accessibility	✗	☑	☑
Audio answering	☑	☑	☑
Real Conversation	☑	☑	☑
Mock Avatar	✗	☑	☑
Storage of user data	☑	✗	☑
Specifying Tracks	☑	✗	☑
Extract user data from resume	✗	✗	☑
Multi-Face Detection	✗	✗	☑
Mobile Detection	✗	✗	☑
Identity Verification	✗	✗	☑

System Modules Algorithms and Models

Facial Emotion Detection :

Algorithm/models	VGGFace	ResNet/50	MobileNetV3
Performance	Strong generalization on facial emotion datasets.	High accuracy and robustness under varying lighting and angles.	Efficient for mobile and real-time applications.
Description	CNN model trained on VGGFace2, fine-tuned for emotion detection on FER2013 or AffectNet.	Deep residual CNN trained for image classification, adaptable for emotion detection.	A compact CNN that performs well for on-device facial emotion recognition with minimal resource usage.
Accuracy	~90–94% on FER2013.	~93% on AffectNet and FER+.	~88–90% on FER2013.
Ease of Use	Moderate — simple fine-tuning required in TensorFlow or PyTorch.	High — pre-trained versions available via PyTorch and Keras.	Very High — optimized for TensorFlow Lite and mobile deployment.

System Modules Algorithms and Models

Eye Tracking:

Algorithm/models	MediaPipe Face Mesh	OpenFace Gaze Estimation	GazeNet (Eye Tracking CNN model)
Performance	Extremely fast (real-time even on mobile). Lightweight and optimized for on-device tracking.	Good performance with head-pose + gaze estimation. Works well in normal lighting.	High performance for accurate gaze prediction but slower than MediaPipe. Works best on GPU.
Description	Lightweight model that detects 468 facial landmarks, enabling fast and accurate eye-tracking on mobile and desktop.	Open-source toolkit that estimates gaze direction using facial landmarks and head pose. Reliable for general eye-tracking tasks.	Deep CNN model trained on gaze datasets to predict eye direction with high accuracy from cropped eye images..
Accuracy	~90–95% landmark accuracy depending on lighting and camera quality.	~85–92% gaze estimation accuracy in standard datasets.	~92–96% accuracy on MPIIGaze / GazeCapture datasets.
Ease of Use	**Very High** — available in Python, C++, and MediaPipe Tasks. Easiest for mobile/web.	**Moderate** — requires installation of OpenFace + dependencies; needs calibration.	**Moderate to Low** — requires deep learning setup (TensorFlow/PyTorch) and dataset for fine-tuning..

System Modules Algorithms and Models

Tone Analysis :

Algorithm/models	Wav2Vac 2.0 (Meta Ai)	Open SMILE (By audEERING)	TRILL (By Google)
Performance	Excellent real-time voice embedding extraction and noise robustness	Good for feature extraction; works well for tone and emotion recognition in speech.	Lightweight and fast; well-suited for mobile/edge tone detection.
Description	A self-supervised model trained on raw audio to learn speech representations; can be fine-tuned for tone or emotion classification.	An open-source toolkit that extracts acoustic features (pitch, loudness, MFCCs) and feeds them into classifiers for tone analysis.	Generates embeddings from short speech segments, enabling emotional tone recognition and voice similarity comparison.
Accuracy	~92% on emotion datasets (RAVDESS, CREMA-D).	~85–88% depending on dataset and classifier used.	~89% on standard emotional speech datasets.
Ease of Use	High — available via Hugging Face and PyTorch with ready pretrained weights.	Moderate — requires manual feature engineering and classifier setup.	High — directly supported in TensorFlow Hub.

System Modules Algorithms and Models

Speech to Text :

Algorithm/models	Whisper (OpenAI)	Deep Speech (Mozilla)	Google Speech-to-Text Api
Performance	Excellent transcription in multiple languages and noisy environments.	Reliable offline speech recognition model.	Excellent real-time transcription with multilingual support.
Description	Transformer-based speech recognition model trained on 680k hours of diverse data.	RNN-based speech-to-text model trained on large English datasets.	Cloud-based API for speech recognition, supporting streaming and batch modes.
Accuracy	WER (Word Error Rate) 5–8%.	~90–93% transcription accuracy.	~95% for high-quality audio.
Ease of Use	High — available through OpenAI API and open -source implementations.	Moderate — easy integration with Python; open-source but requires preprocessing.	Very High — simple REST API integration; ideal for production systems.

System Modules Algorithms and Models

Text to Speech :

Algorithm/models	Piper (en_US-lessac-medium)	Tacotron 2	FastSpeech
Performance	Extremely fast with real-time inference on CPU. Optimized for offline and low-resource environments.	High-quality output but slower due to its autoregressive architecture. Not suitable for real-time applications.	Very fast and efficient. Uses a non-autoregressive approach, enabling near real-time synthesis.
Description	Lightweight neural TTS engine using pre-trained ONNX models. Designed for offline usage with low latency and minimal dependencies.	Sequence-to-sequence model with attention that converts text into mel-spectrograms, followed by a vocoder (e.g., WaveNet).	Non-autoregressive TTS model that generates speech in parallel, improving speed and stability over Tacotron 2.
Accuracy	Good speech quality and intelligibility. Slightly less expressive than larger deep learning models.	Very high naturalness and human-like speech quality. Considered one of the most realistic TTS models.	High naturalness with stable and consistent output. Slightly less expressive than Tacotron 2..
Ease of Use	**Very High** — simple setup using ONNX, works fully offline, and easy to integrate into applications	**Moderate** — requires complex setup (Tacotron + vocoder) and typically needs GPU support.	**Moderate** — easier than Tacotron 2 but still requires model setup or use of pre-trained models.

System Modules Algorithms and Models

Multi-Face Detection :

Algorithm/models	OpenCV Haar Cascade	MTCNN (Multi-task CNN)	Dlib (HOG + CNN)
Performance	Very fast and works in real-time on CPU. Suitable for simple applications but less robust in complex scenes.	Moderate speed with good balance between detection and alignment. Slightly slower due to multi-stage processing.	HOG is fast on CPU, CNN is slower but more accurate. Overall moderate performance.
Description	Traditional machine learning-based method using Haar-like features and cascade classifiers for face detection.	Deep learning-based model that performs face detection and facial landmark localization in multiple stages.	Combines HOG-based detection (fast) and CNN-based detection (more accurate). Widely used for face recognition pipelines.
Accuracy	~80–88% accuracy. Struggles with different angles, lighting conditions, and occlusions.	~90–96% accuracy. Handles different poses, lighting, and multiple faces effectively.	~88–95% accuracy depending on method (higher with CNN). Good performance in varied conditions.
Ease of Use	**Very High** — simple to use with OpenCV, no training required, lightweight and easy to integrate.	**Moderate** — requires deep learning libraries and pre-trained models. Slightly complex setup.	**Moderate** — requires installation of Dlib and dependencies. Easier than MTCNN but heavier than Haar Cascade.

System Modules Algorithms and Models

Mobile Usage Detection:

Algorithm/models	VGGNet (VGG-based Model)	OpenFace Gaze Estimation	YOLO
Performance	High performance in feature extraction but slower due to deep architecture. Not ideal for real-time mobile applications.	Fast and efficient, optimized for mobile and embedded devices with low latency.	Very high performance for real-time object detection. Processes images in a single pass, making it extremely fast.
Description	Deep CNN model with multiple convolutional layers. Known for its simplicity and strong feature representation, often used in classification tasks.	Lightweight CNN architecture using depthwise separable convolutions, designed specifically for mobile and low-power devices.	Real-time object detection model that divides the image into grids and predicts bounding boxes and class probabilities in one step.
Accuracy	High accuracy in classification tasks (~90–93%) but less efficient for detection unless combined with other frameworks.	Good accuracy (~85–92%) with a trade-off between speed and performance.	Very high accuracy (~90–96%) depending on the version (YOLOv5, YOLOv8, etc.) and dataset.
Ease of Use	**Moderate** — requires more computational resources and is less optimized for mobile deployment.	**Very High** — easy to deploy on mobile using TensorFlow Lite or similar frameworks.	**Moderate** — requires setup and understanding of detection pipelines but widely supported with pre-trained models.

System Modules Algorithms and Models

Identity Verification :

Algorithm/models	ArcFace	FaceNet	VGGFace2
Performance	Extremely fast with optimized backbones (e.g., MobileNet). Suitable for real-time and edge devices.	Highly efficient for generating face embeddings. Scalable and fast for matching.	Moderate speed due to deeper standard architectures (e.g., ResNet-50). Balanced performance.
Description	Deep CNN employing Additive Angular Margin Loss to maximize class distinction and reduce variance.	Direct mapping of face images into compact Euclidean space using Triplet Loss for similarity measuring.	Deep CNN model trained on massive, diverse datasets for robustness to pose and lighting.
Accuracy	State-of-the-art accuracy (e.g., LFW 99.8%+). Strong performance in varied conditions.	Very high accuracy (e.g., LFW 99.6%+). Good general performance in standard scenarios.	High accuracy across diverse populations. Proven robust model but generally outperformed by newer SOTA.
Ease of Use	Moderate — Requires precise pre-processing and model selection, but pre-trained models exist.	High — Easier to use with widespread framework support and multiple open-source versions.	High — Simple to use as standard ResNet feature extractor, with pre-trained weights widely available.

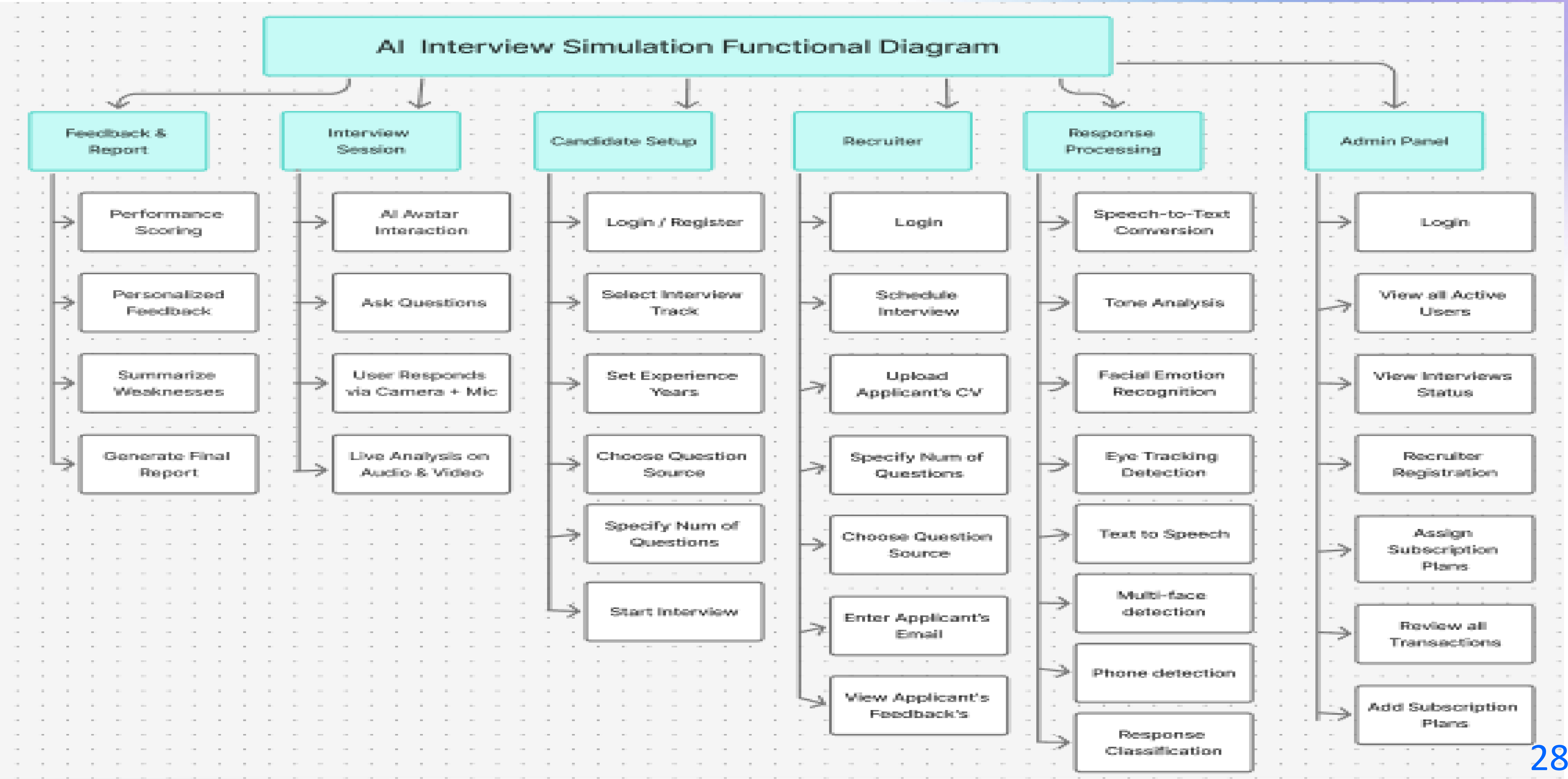
Proposed System

Functional Diagram(FD)

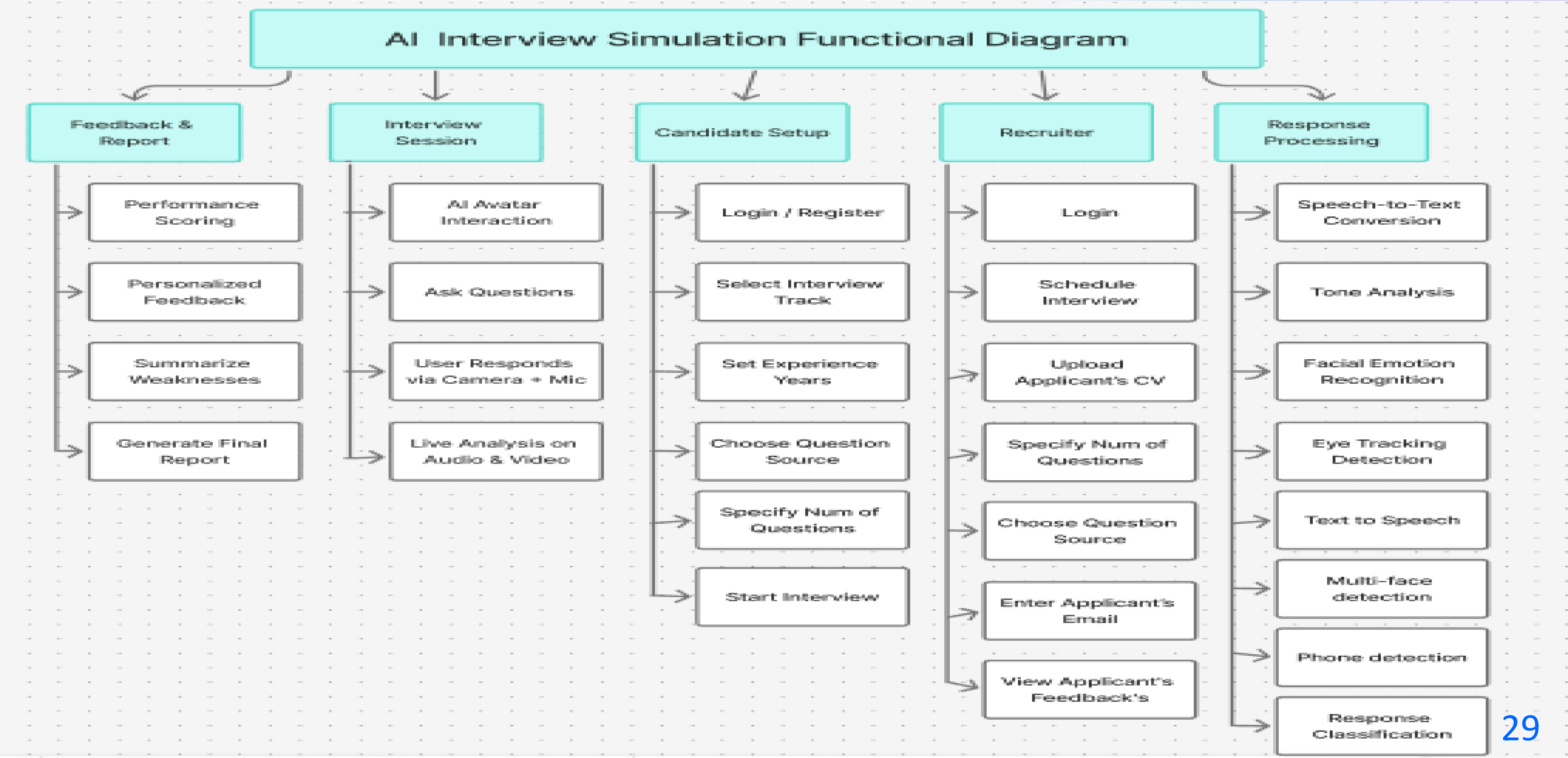
Entity Relationship Diagram(ERD)

Object Oriented Modelling(OOM)

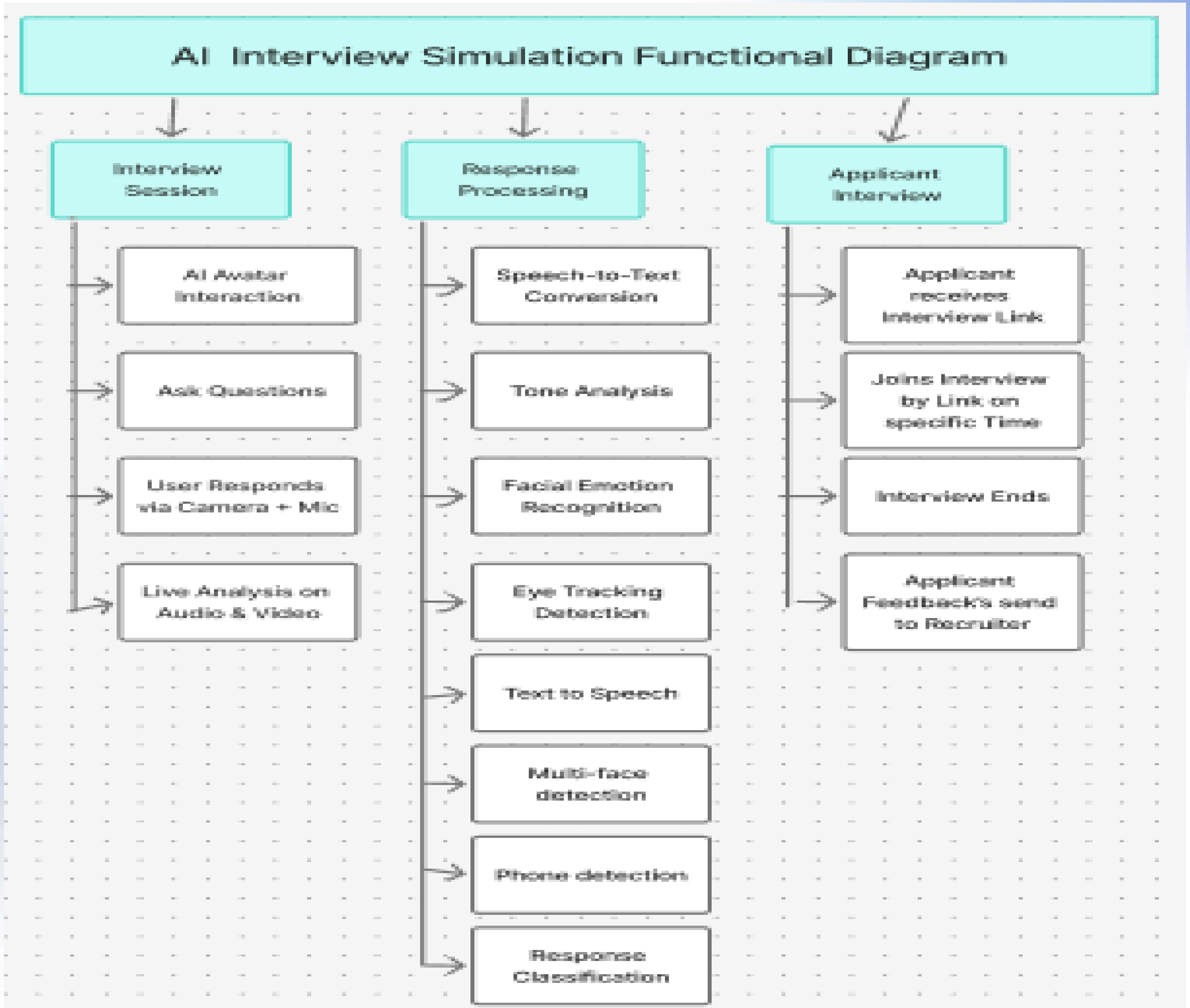
Web Application Functional Diagram



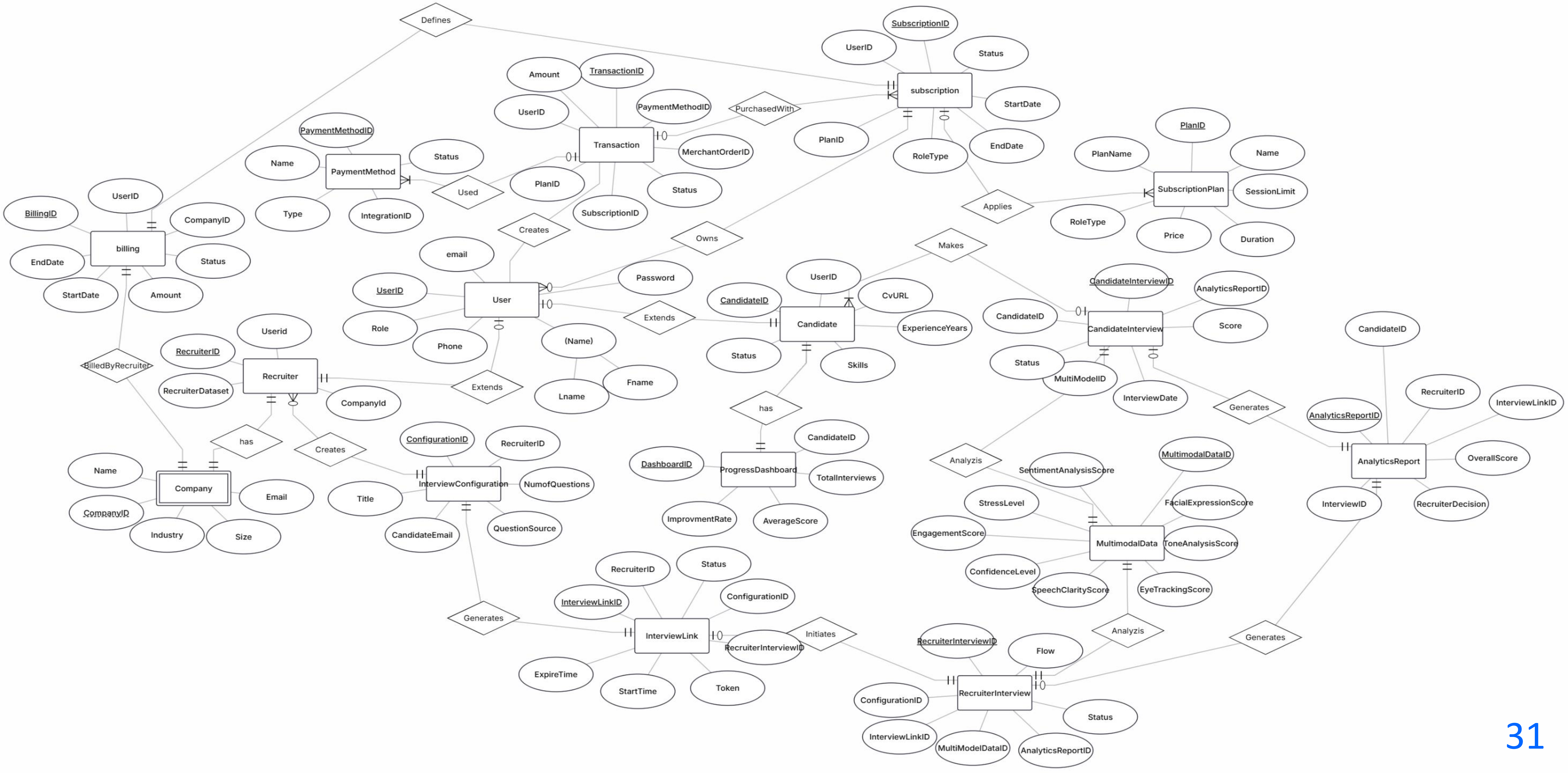
Mobile Application Functional Diagram



Desktop Application Functional Diagram



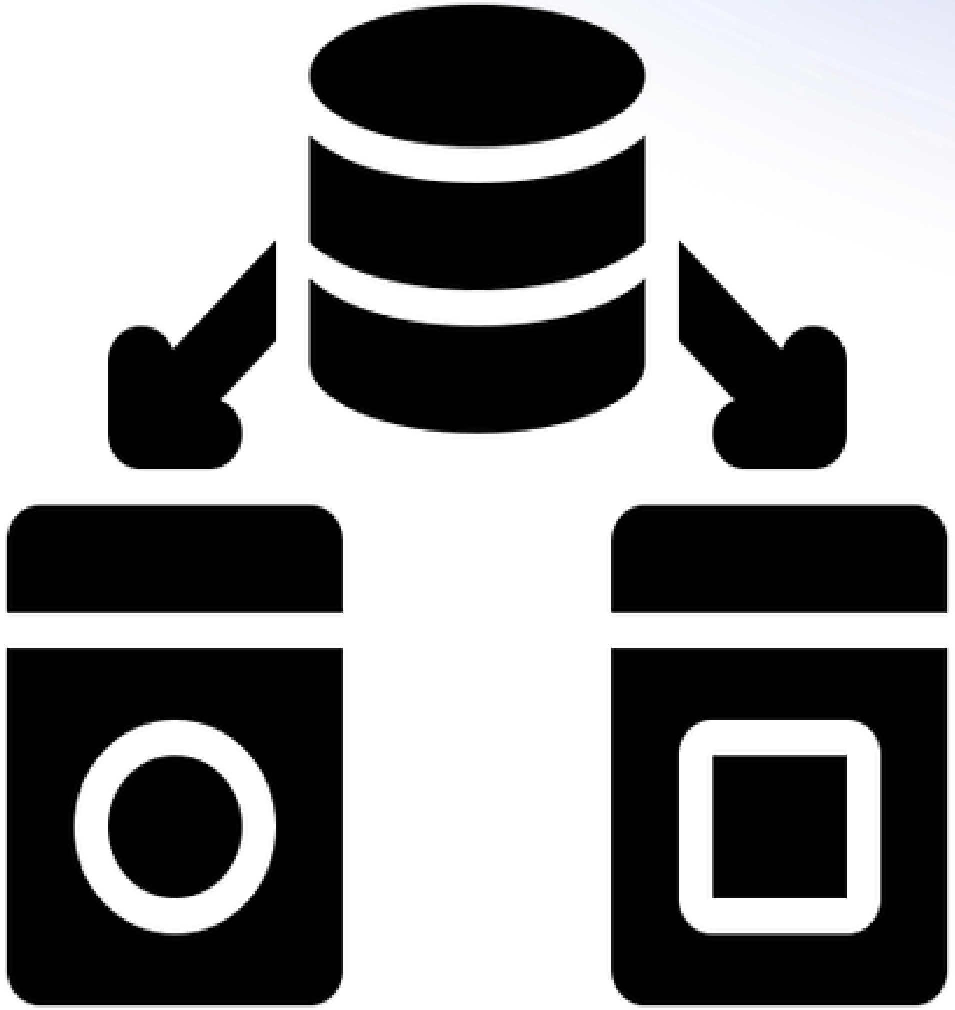
Entity Relationship Diagram (ERD)



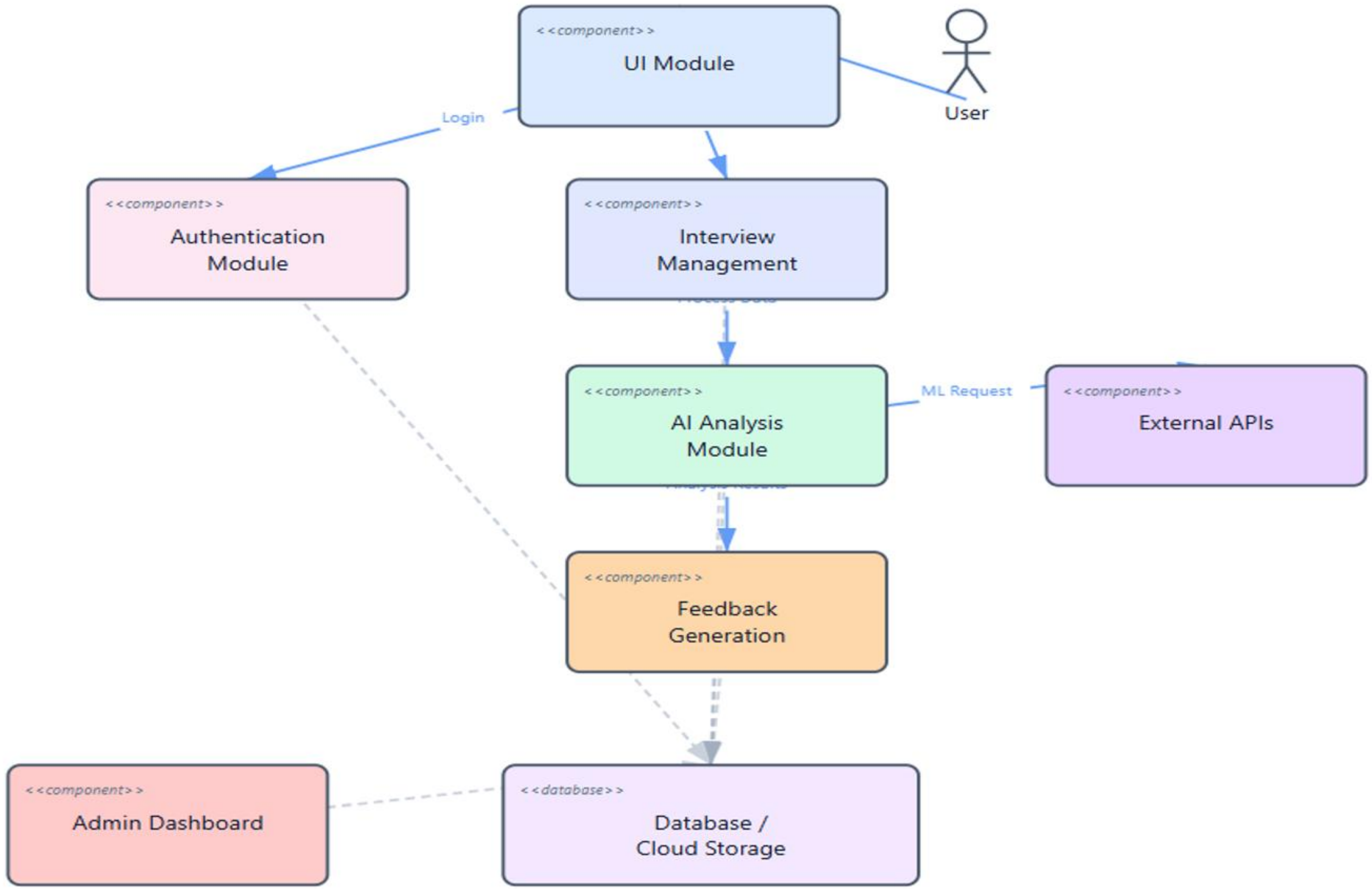
Object-oriented Modelling (OOM)

9 Diagrams :

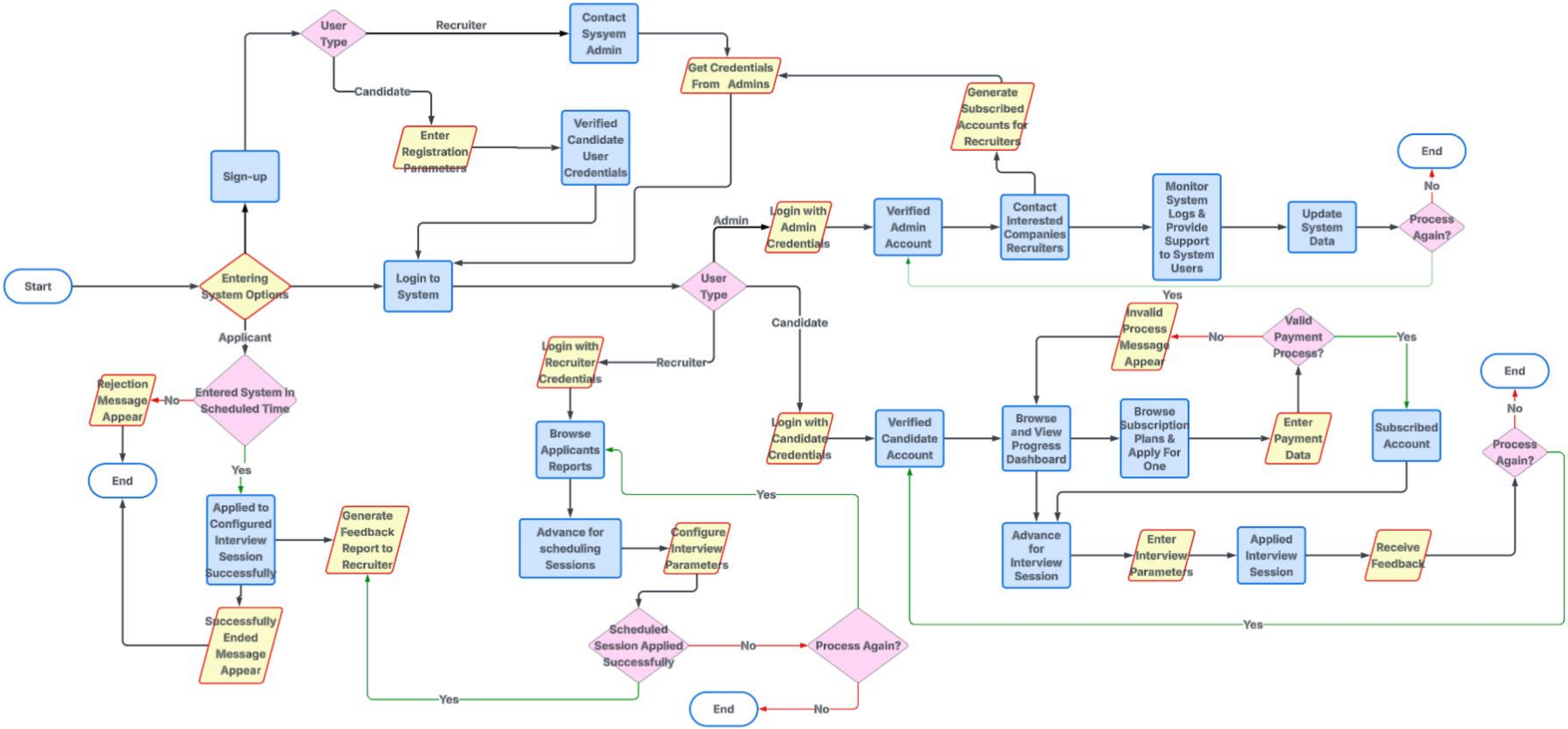
- Use case diagram
- Class diagram
- Object diagram
- Sequence Diagram
- Activity Diagram
- Components Diagram
- package Diagram
- Timing Diagram
- Communication Diagram



Communication Diagram



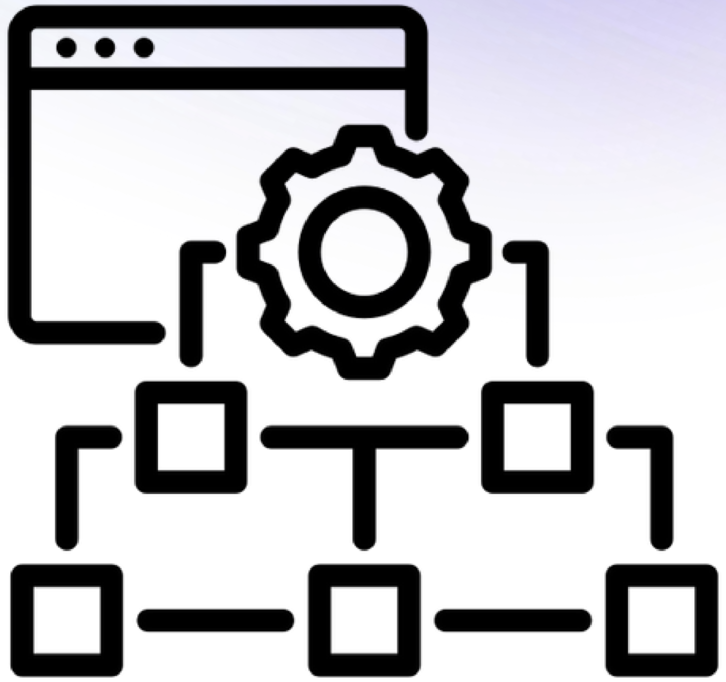
Flow Chart Diagram



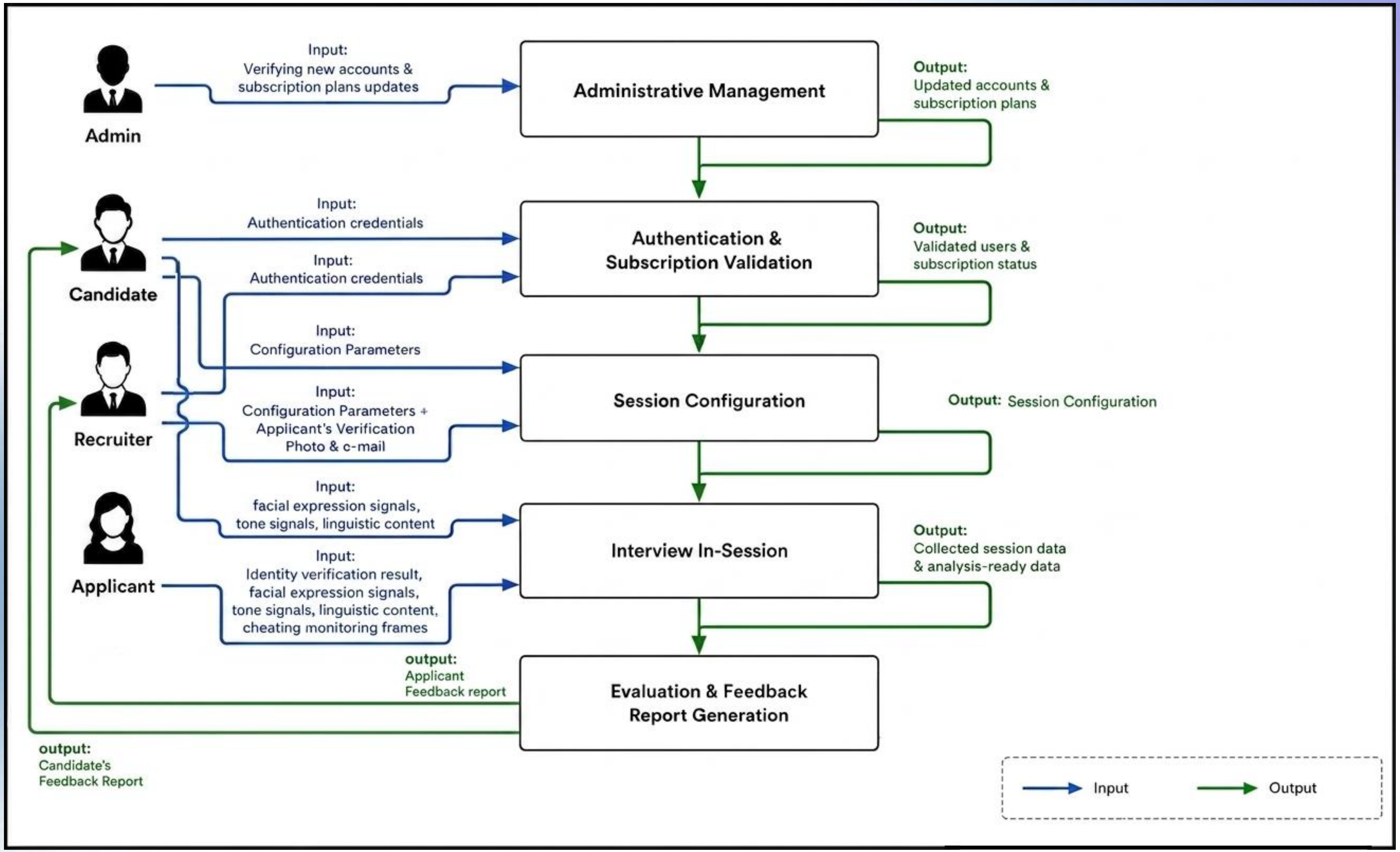
System Architecture & Design

System Architecture & Design

- **Overall Architecture**
- **Proposed Algorithms**



Overall System Block Diagram Architecture



Proposed Algorithms

Proposed Facial Expression Detection Model

Why choose VGGFace?

High-Quality Feature Extraction:
Pretrained on a massive face dataset, giving strong understanding of facial features.

Excellent Accuracy for Facial Tasks:
Its deep CNN architecture captures subtle emotional expressions very well.

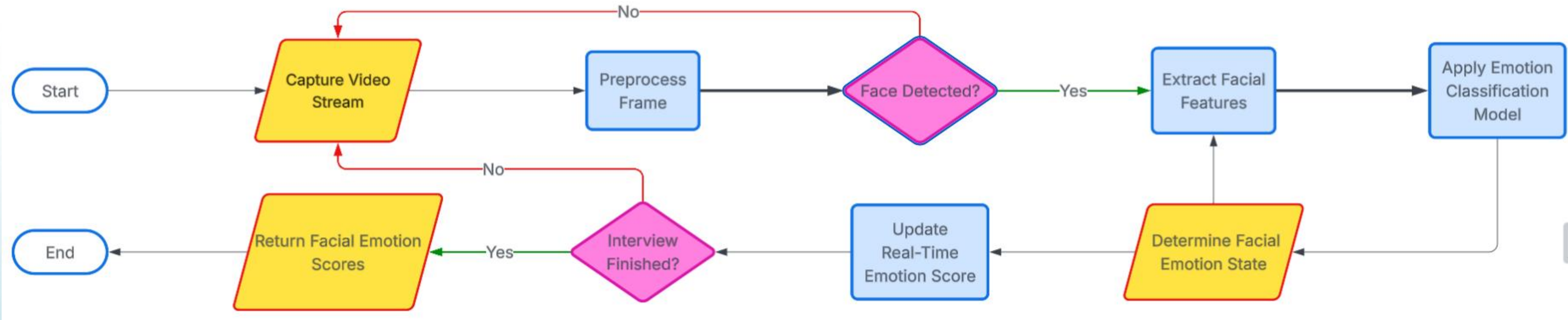
Transfer Learning Friendly:
Easy to fine-tune for emotion classification with minimal data.

Robust Against Variations:
Performs well under different lighting, angles, and face shapes.

Mature & Well-Supported Model:
Widely used in academia and industry, ensuring reliability.

Model Flowchart

Facial Expressions Detection



Proposed Tone Analysis Model

Why choose Wav2Vec2?

High-Quality Vocal Feature Extraction:
Learns rich speech representations that capture tone, pitch, rhythm, and speaking patterns.

Excellent Accuracy for Tone Recognition:
Identifies vocal cues such as confidence and engagement.

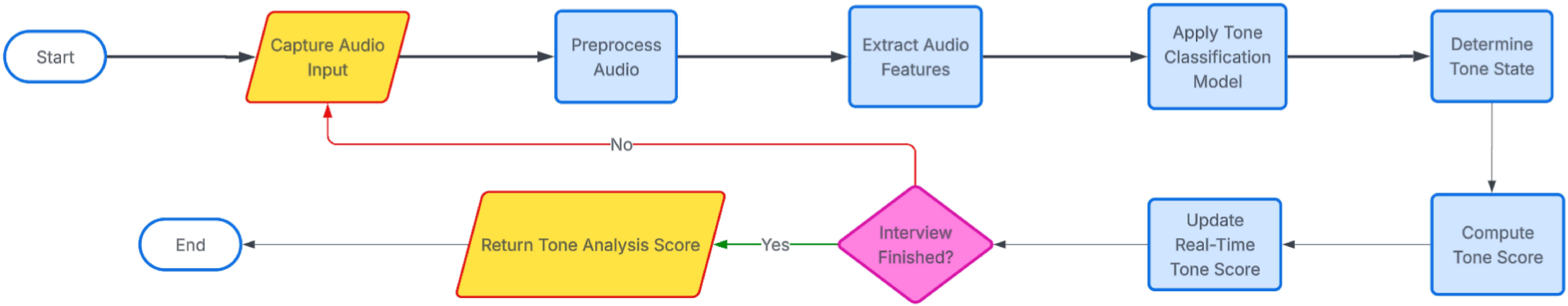
Transfer Learning Friendly:
Can be fine-tuned with limited task-specific data

Robust Against Speaker Variations:
Performs well across different voices, accents, and speaking styles..

State-of-the-Art Speech Model:
Widely used for speech understanding and emotion analysis.

Model Flowchart

Tone Analysis



Proposed Eye Tracking Model

Why choose MediaPipe Face Mesh?

Real-Time Performance:
Lightweight and optimized to run fast even on CPU.

High Landmark Precision:
Provides 468 facial landmarks, including detailed eye contours.

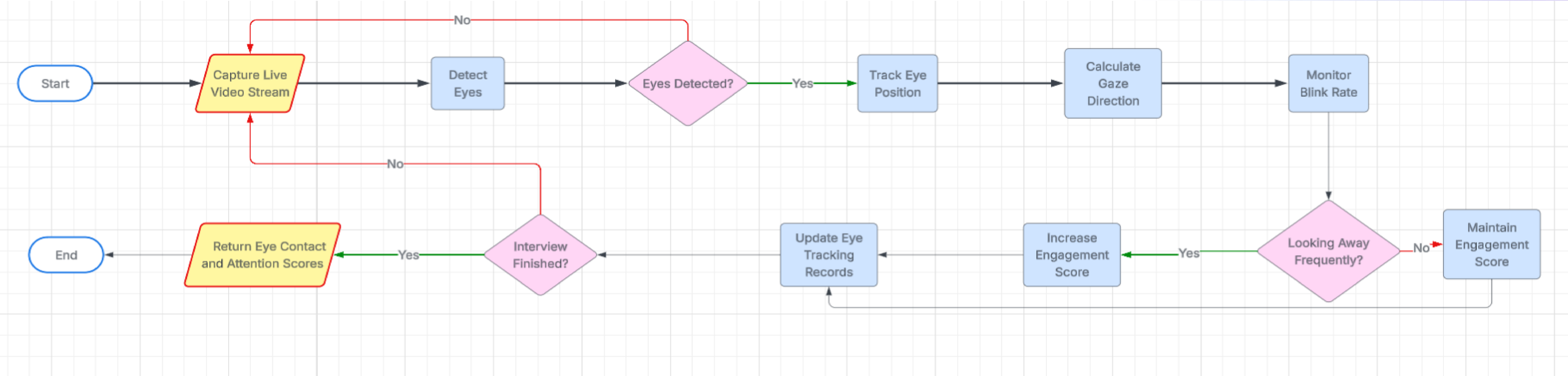
Cross-Platform Compatibility:
Works in Python, Android, iOS, and Web.

Easy Integration:
Simple API with ready-to-use face mesh models.

Efficient & Low Resource Usage:
Perfect for live tracking without heavy computational cost.

Model Flowchart

Eye Tracking



Proposed Generating Questions Model

Why choose deepseek?

Strong Language Understanding:
Known for generating structured and accurate content.

High Reasoning Ability:
Good at producing challenging, varied questions.

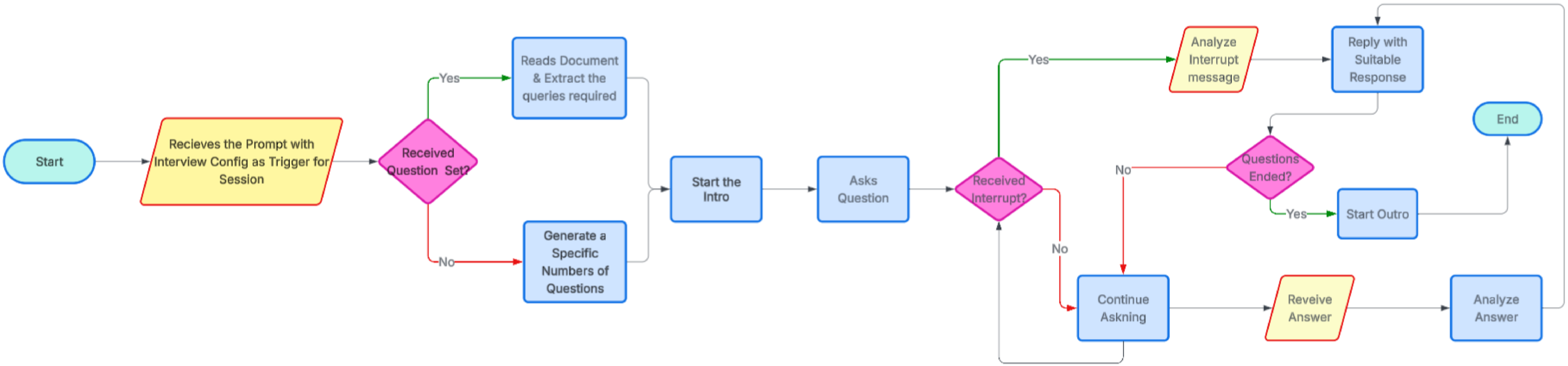
Efficient and Fast:
Generates results quickly even for complex prompts.

Stable Output Quality:
Consistent tone, logic, and formatting.

Cost-Effective Model:
Power of advanced LLMs at relatively lower resource requirements.

Model Flowchart

Deepseek LLM



Proposed Speech-To-Text Model

Why choose whisper?

State-of-the-Art Accuracy:
Excellent transcription for multiple languages and accents.

Robust to Noise:
Performs well even in noisy environments.

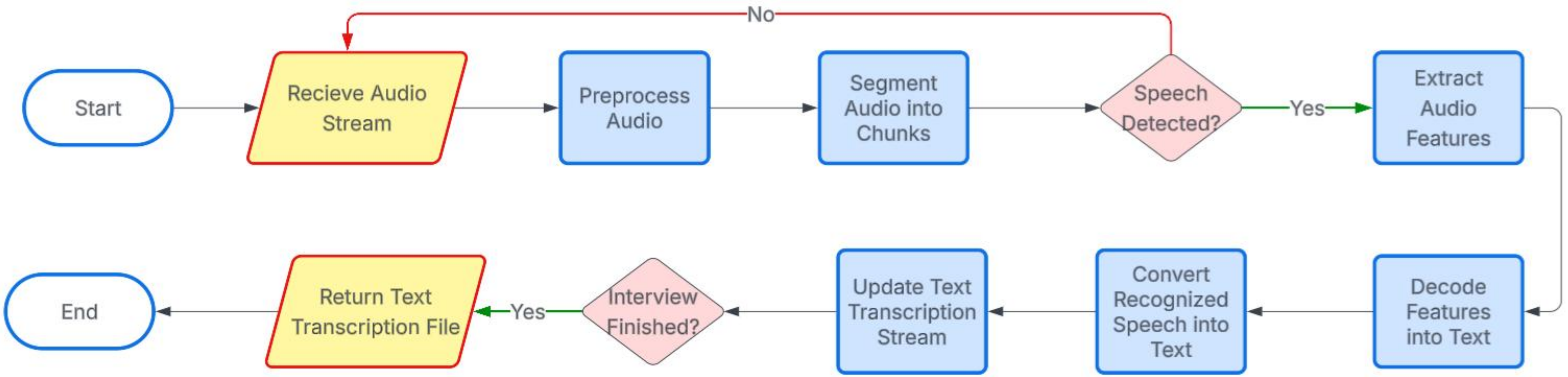
Open-Source and Free:
Full access without licensing costs.

Multilingual Support:
Handles dozens of languages natively.

Strong Timestamp & Segment Detection:
Great for aligning speech with avatar animation or logs.

Model Flowchart

Speech-To-Text



Proposed Text-to-Speech Model

Why choose Piper?

High-Quality Speech Processing:
Learns deep speech patterns for natural audio interaction.

Real-Time Conversation Capability:
Efficient for live interview communication and response generation.

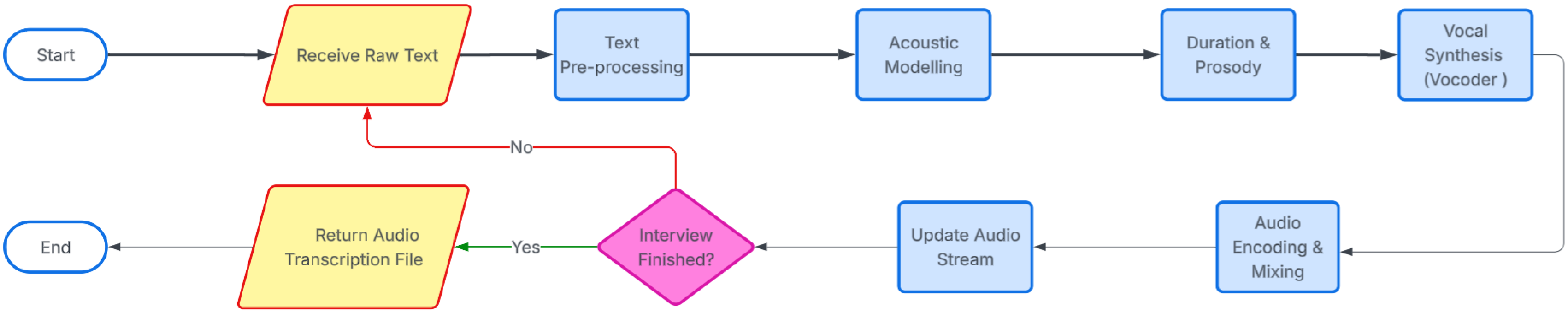
Robust Audio Understanding:
Handles different accents, tones, and speaking styles effectively.

Transformer-Based Architecture:
Provides advanced contextual understanding for human-like speech generation.

Open-Source & Research Proven:
Widely adopted in modern speech AI applications.

Model Flowchart

Text-To-Speech



Proposed Mock Avatar

Why choose HeyGen to get the Avatar?

High-Realism Avatars:
Produces extremely natural facial and lip-sync animations.

API Availability:
Ready-to-use from the browser without complicated setup.

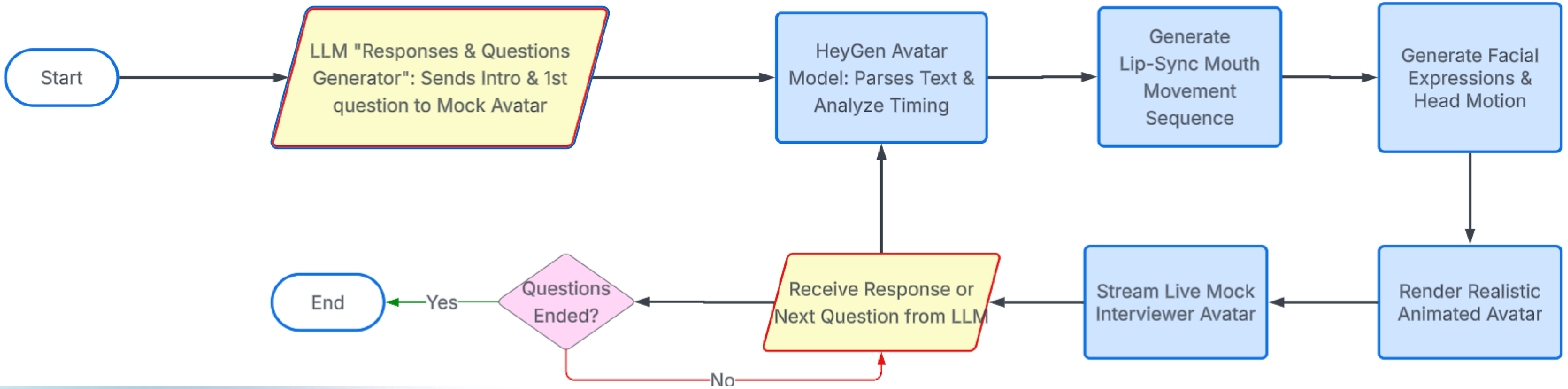
Fast Video Generation:
Generates speaking avatars in minutes.

Flexible Customization:
Supports changing voice, style, language, and persona.

Reliable Cloud Service:
Offloads heavy rendering to HeyGen servers..

Model Flowchart

Mock Interviewer Avatar



Proposed Multi-Face detection Model

Why choose Haar Cascade?

Fast Real-Time Detection:
Lightweight and optimized for instant face detection during live sessions.

Low Computational Cost:
Runs efficiently on devices with limited hardware resources.

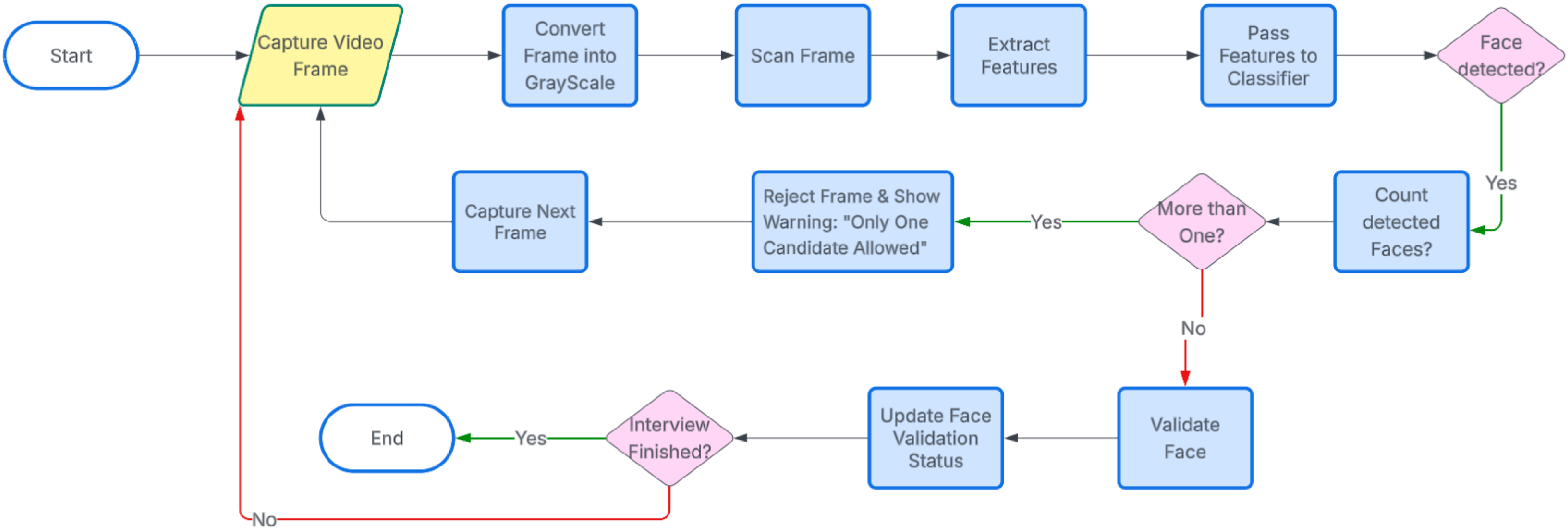
Reliable Multi-Face Detection:
Accurately detects multiple faces within interview environments.

Easy OpenCV Integration:
Simple to implement and widely supported in computer vision systems.

Suitable for Security Monitoring:
Helps ensure only the intended candidate appears during the interview.

Model Flowchart

Multi-Face Detection



Proposed Mobile Phone detection Model

Why choose VGGNet?

Powerful Feature Extraction:
Captures detailed visual patterns for accurate phone detection.

High Detection Accuracy:
Performs strongly in object recognition and classification tasks.

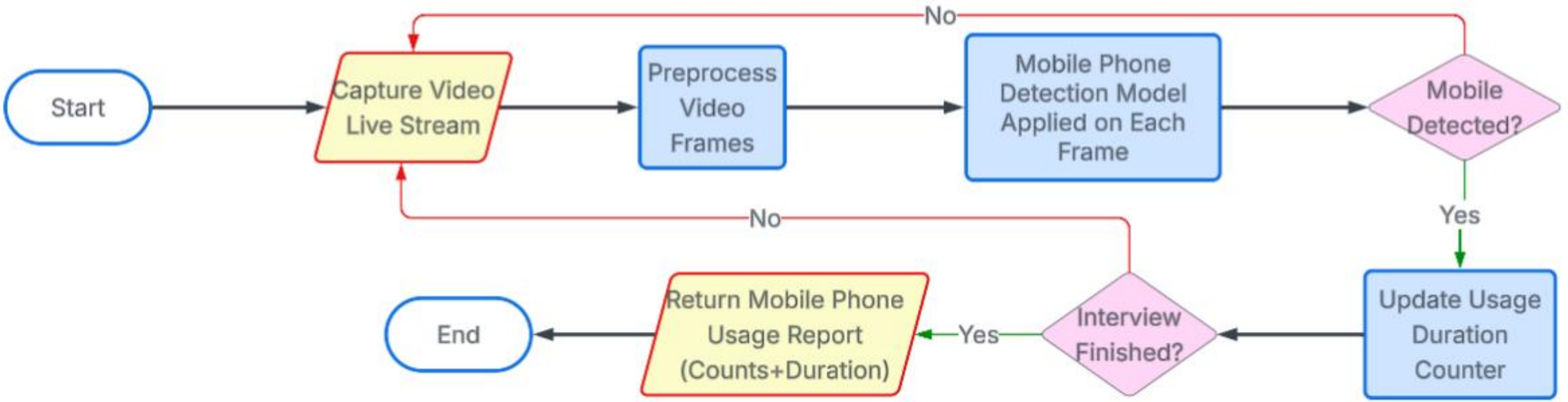
Deep CNN Architecture:
Learns complex phone shapes and visual variations effectively.

Transfer Learning Friendly:
Can be fine-tuned efficiently using custom datasets.

Reliable & Research Proven:
One of the most trusted CNN models in computer vision.

Model Flowchart

Mobile Phone Usage Detection



Proposed Identity Verification Model

Why choose ArcFace?

High Recognition Accuracy:
Provides strong facial identity verification and matching performance.

Deep Facial Feature Embedding:
Captures unique facial characteristics accurately using deep learning.

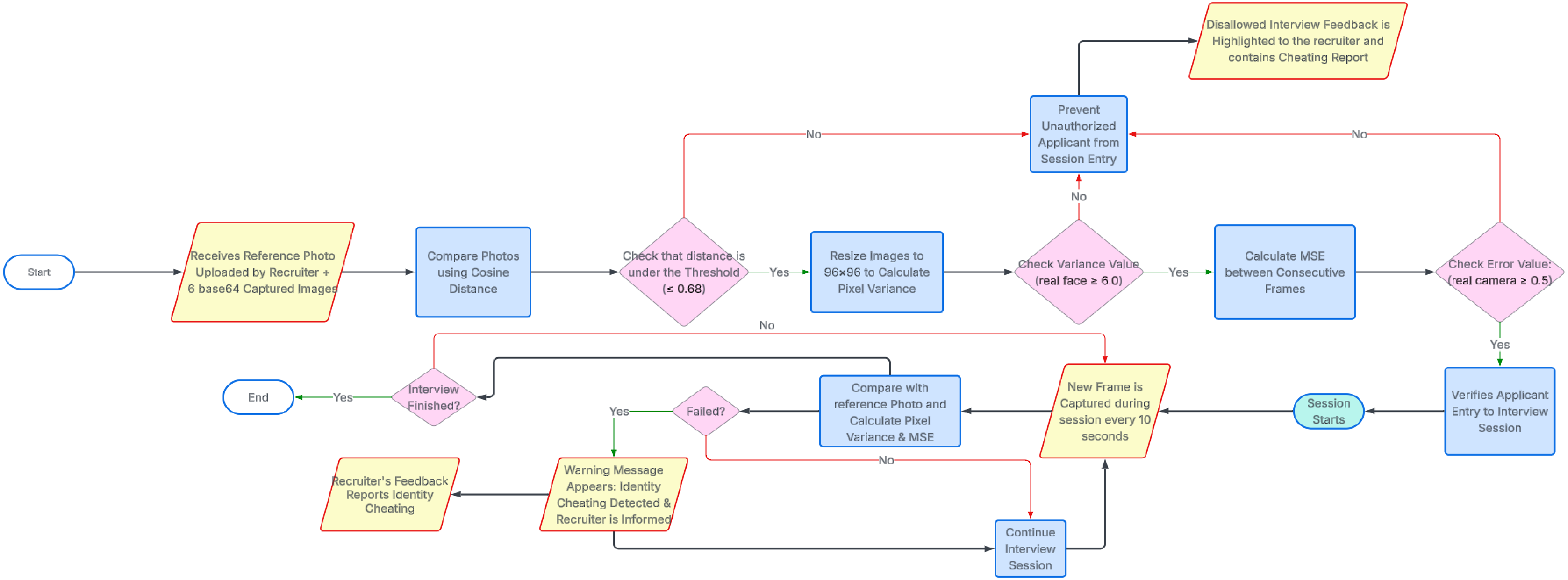
Robust Against Variations:
Handles different lighting conditions, angles, and facial expressions effectively.

Real-Time Verification Capability:
Suitable for live interview identity authentication.

Trusted & Research Proven:
Widely used in facial recognition and identity verification systems.

Model Flowchart

Identity Verification



The Evaluation Algorithm :

The Evaluation Module generates a comprehensive interview assessment by collecting, analyzing, and combining multiple AI-driven evaluation models into one unified scoring system.
Each model measures a different aspect of candidate performance, and all generated scores are fused using weighted mathematical formulas to produce a balanced, fair, and accurate final evaluation.

Behavioral & Communication Analysis

- Facial Expression Analysis Eye Tracking
- Analysis Tone & Voice Analysis
- Confidence & Engagement Detection
- Stress & Emotional Response Monitoring

Knowledge & Technical Evaluation

- Answer Accuracy Evaluation
- Role-Specific Question Assessment
- Technical Knowledge Scoring
- Response Quality & Relevance Evaluation

Security & Anti-Cheating Monitoring

- Multi-Face Detection
- Identity Verification
- Mobile Phone Detection
- Candidate Presence Validation
- Eye Tracking Analysis

The Evaluation Algorithm (Cont.) :

Evaluation Score Fusion & Reporting

All generated scores, behavioral signals, and interview interactions are combined through a weighted evaluation formula that balances technical performance, communication skills, behavioral indicators, and interview integrity to produce accurate and unbiased interview assessments. The system also generates detailed interview transcripts, personalized improvement recommendations, and performance insights that help candidates better understand their strengths, weaknesses, and overall interview readiness.

For practice sessions, candidates receive personalized feedback reports highlighting strengths, weaknesses, improvement recommendations, and overall interview readiness to support continuous skill development.

For recruiter-managed hiring sessions, evaluation reports are delivered directly to recruiters with additional integrity and attendance monitoring indicators. Identity verification failures or cheating-related detections are automatically flagged and highlighted within the recruiter’s feedback and assessment dashboard to support secure and reliable hiring decisions.

System Implementation

System Implementation (Technologies & Tools)

Hardware Tools Used:

Processing
32/64-bit CPU
8 GB RAM

Webcam
Facial Analysis
Anti-Cheating Monitoring

Microphone
Speech-to-Text
Tone Analysis

Headphones
Noise Reduction
Clear Communication

Software Technologies Used:

The system was implemented as a fully integrated architecture composed of three major layers working in real time: the frontend layer, the backend infrastructure, and the AI engine.

The frontend was implemented using Flutter and Dart to provide a unified cross-platform experience across **web**, **mobile**, and **desktop** environments.

The backend was implemented using Node.js , Express **and** Websocket as the central orchestration layer responsible for managing communication between all frontend platforms, database services, and the AI engine. It acts as the core controller of the entire interview ecosystem.

The AI engine was implemented using FastAPI and multiple integrated AI models responsible for conducting and evaluating interview sessions in real time.

System Implementation (Cont.)

Frontend Platform-Specific Roles :

Web App: Main control hub for candidates, recruiters & admins — scheduling, uploads, reports, and platform management.

Mobile App: Simplified, candidate-focused access to reports, progress, profiles, CV uploads & subscriptions on the go

Desktop App: Powers real hiring interviews — live AI sessions with identity verification, anti-cheating & behavioral analysis.

Backend : Built on Node.js & Express, the backend is the central orchestration layer — connecting every frontend platform, the database, and the AI engine.

Interview & Data Management
Scheduling, CV processing, reports & storage

System Services
Subscriptions, payments, email, admin ops

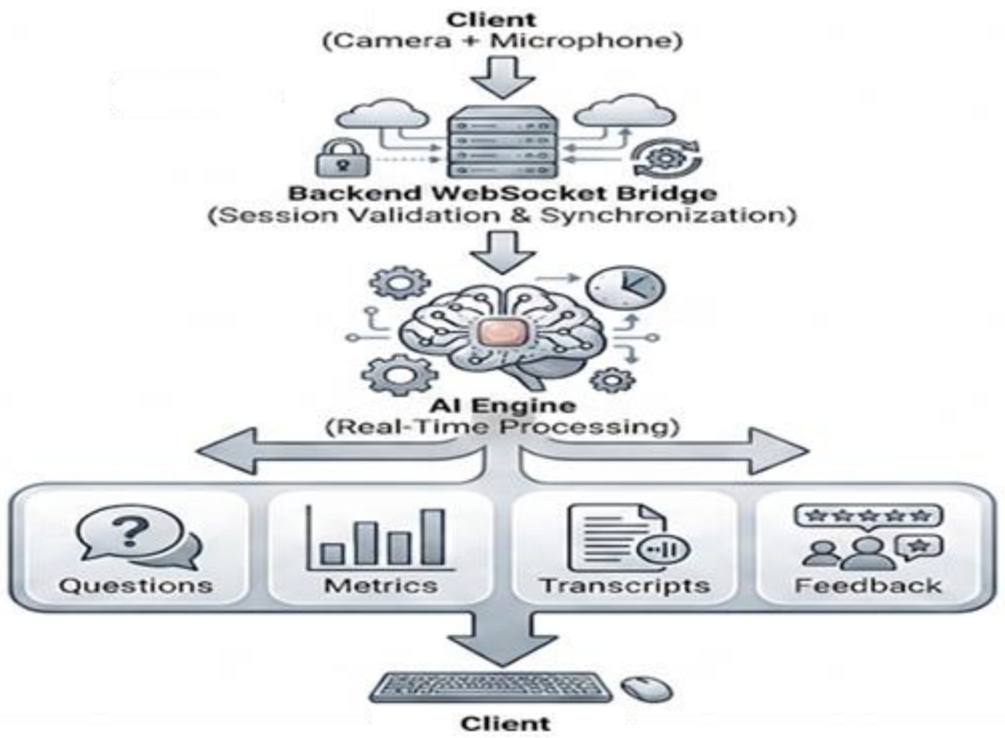
Authentication & Security
JWT, role-based access, secure sessions

One of the most important responsibilities of the backend is managing real-time communication between the interview client and the AI engine through a **dedicated WebSocket bridge** architecture.

System Implementation (Cont.)

Backend WebSocket Responsibilities

- Session Validation
- AI Session Initialization
- Media Stream Forwarding
- State Synchronization
- Reconnection Handling
- Event Persistence



AI Responses Through the Bridge

- Generated Questions
- Behavioral Metrics
- Transcript Updates
- Integrity Signals
- Final Evaluation Results

The AI Engine Role: Unlike traditional interview systems, the AI engine continuously processes live video frames, audio streams, behavioral signals, and interview context throughout the entire interview session.

Visual & Behavioral Analysis Models

Facial expression recognition	VGG Facial Model
Eye tracking and face landmark detection	MediaPipe
Phone and object detection	SSD300-VGG16
Face counting and candidate presence validation	Haar Cascade

Conversational & Audio Intelligence Models

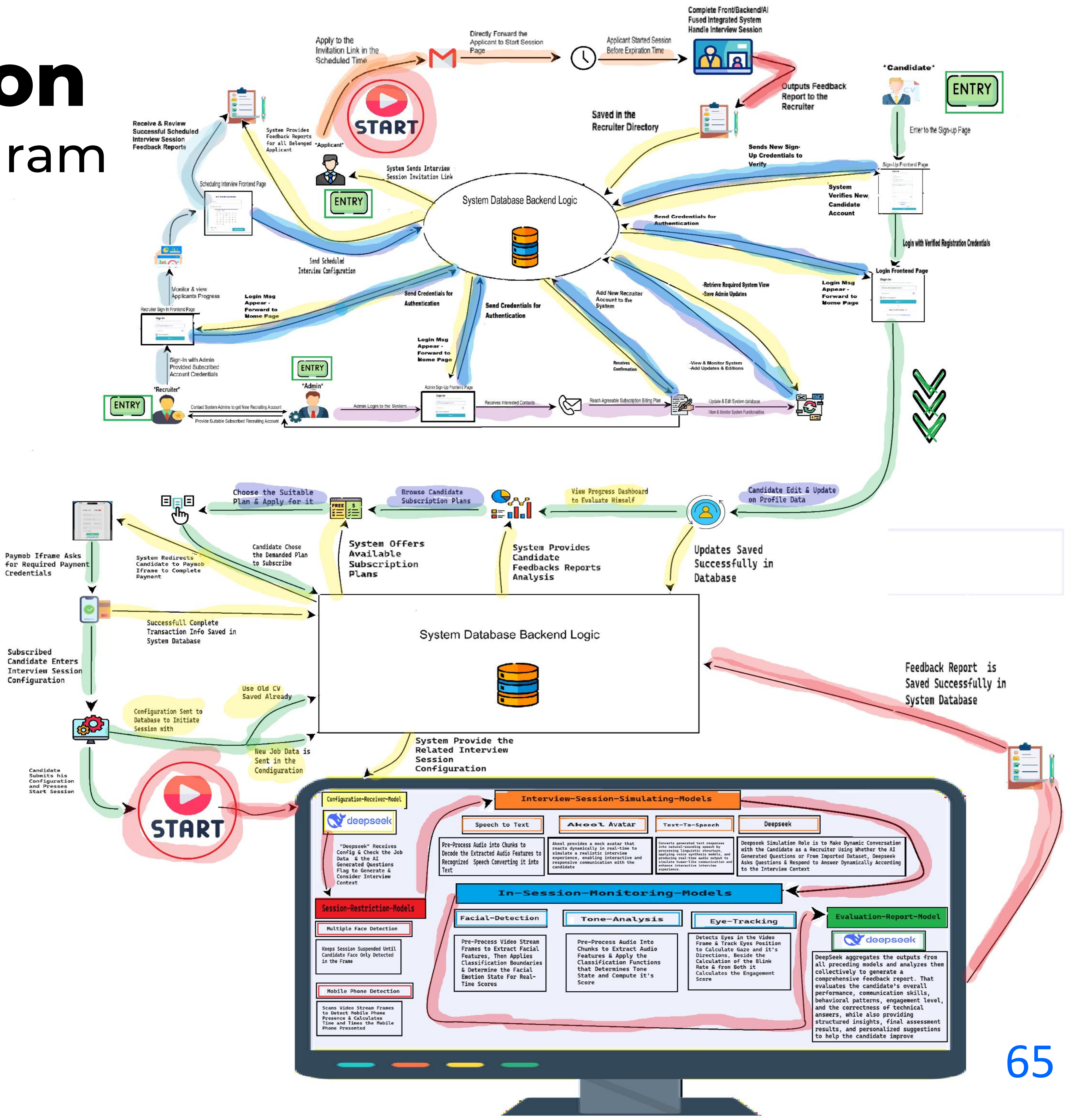
Vocal tone and audio feature analysis	Wav2Vec
Speech transcription	Whisper
Question generation and feedback generation	DeepSeek LLM
AI-generated interviewer speech	Piper TTS
Realistic interviewer animation and lip-sync interaction	HeyGen Avatar

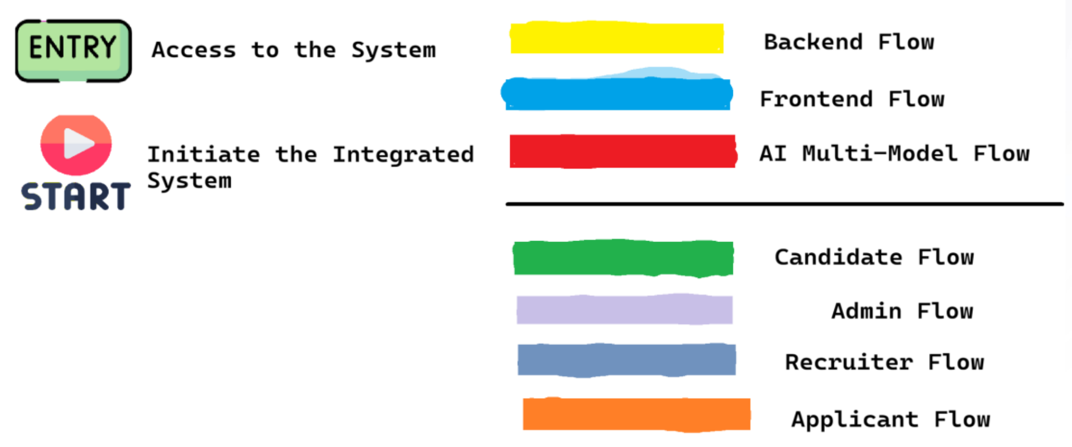
deepseek MediaPipe Whisper VGG OpenCV piper HeyGen

System Implementation

Integrated System Data-Flow Diagram

- Shows how interaction with the system begins, including entry flow and required pre-processing steps before reaching the AI environment.
- Illustrates how system components (frontend, backend, and AI) are connected and work together.
- Highlights the integration of AI models for simulation, monitoring, and evaluation within the system.
- Demonstrates the data flow throughout the session, ending with report generation and storage.





System Implementation Testing Results

Objective

Validate that the integrated AI assessment framework produces distinguishable overall feedback across diverse interview behaviors, performance levels, and policy-compliance conditions.

No.	Scenario	Overall Score
1	The candidate demonstrates outstanding behavioral, vocal, technical, and non-verbal performance with no mobile phone usage or multiple-face violations detected.	92%
2	The candidate performs well in behavioral, vocal, and non-verbal aspects but achieves only an average technical score, with no violations detected.	74%
3	The candidate demonstrates excellent technical competency but receives lower behavioral, vocal, and non-verbal scores, while maintaining full compliance with interview policies.	65%
4	The candidate performs adequately in behavioral, vocal, and technical assessments but exhibits weak eye contact, facial engagement, and overall non-verbal communication.	70%
5	The candidate achieves satisfactory scores across all evaluation dimensions; however, mobile phone usage is detected during the interview session, reducing the overall assessment score.	58%
6	The candidate performs well in all assessment categories, but the system detects the presence of additional individuals during the interview session.	62%
7	The candidate obtains acceptable performance scores but triggers both mobile phone usage and multiple-face detection violations during the interview.	48%
8	The candidate receives low behavioral, vocal, technical, and non-verbal scores, accompanied by multiple interview policy violations, resulting in a significantly reduced overall score.	28%
9	Successful Identity Verification	-
10	Failed Identity Verification	-

Testing Coverage

High Performance Scenarios

- Excellent performance across all assessment dimensions
- Strong communication with moderate technical competency
- Strong technical competency with weaker soft skills

Non-Verbal Performance Scenarios

- Adequate verbal and technical performance with weak eye contact, facial engagement, and body language

Policy Violation Scenarios

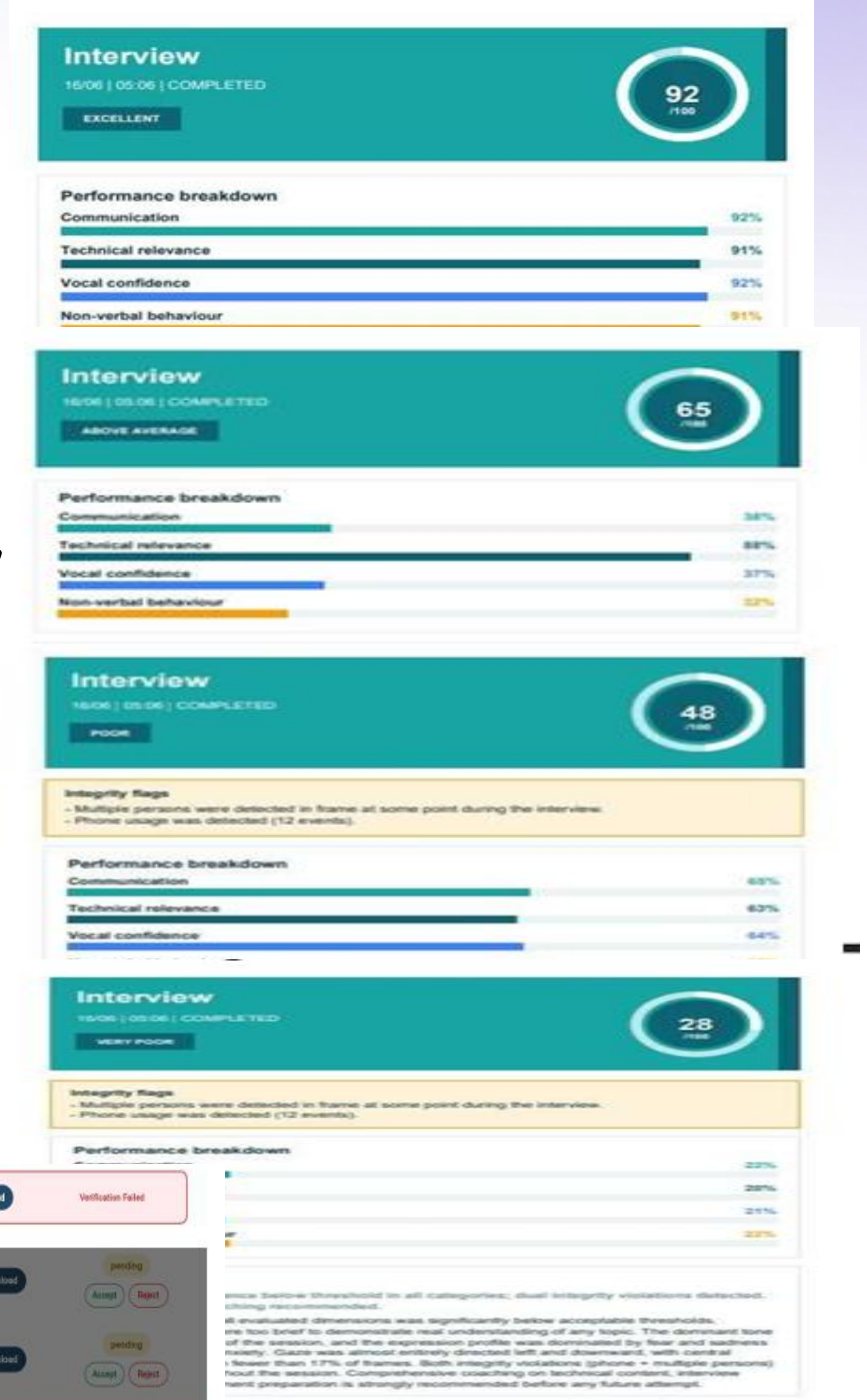
- Mobile phone usage detected during the interview
- Multiple-face detection violation
- Combined mobile phone and multiple-face violations

Low Performance Scenarios

- Weak behavioral, vocal, technical, and non-verbal performance wit additional policy violations

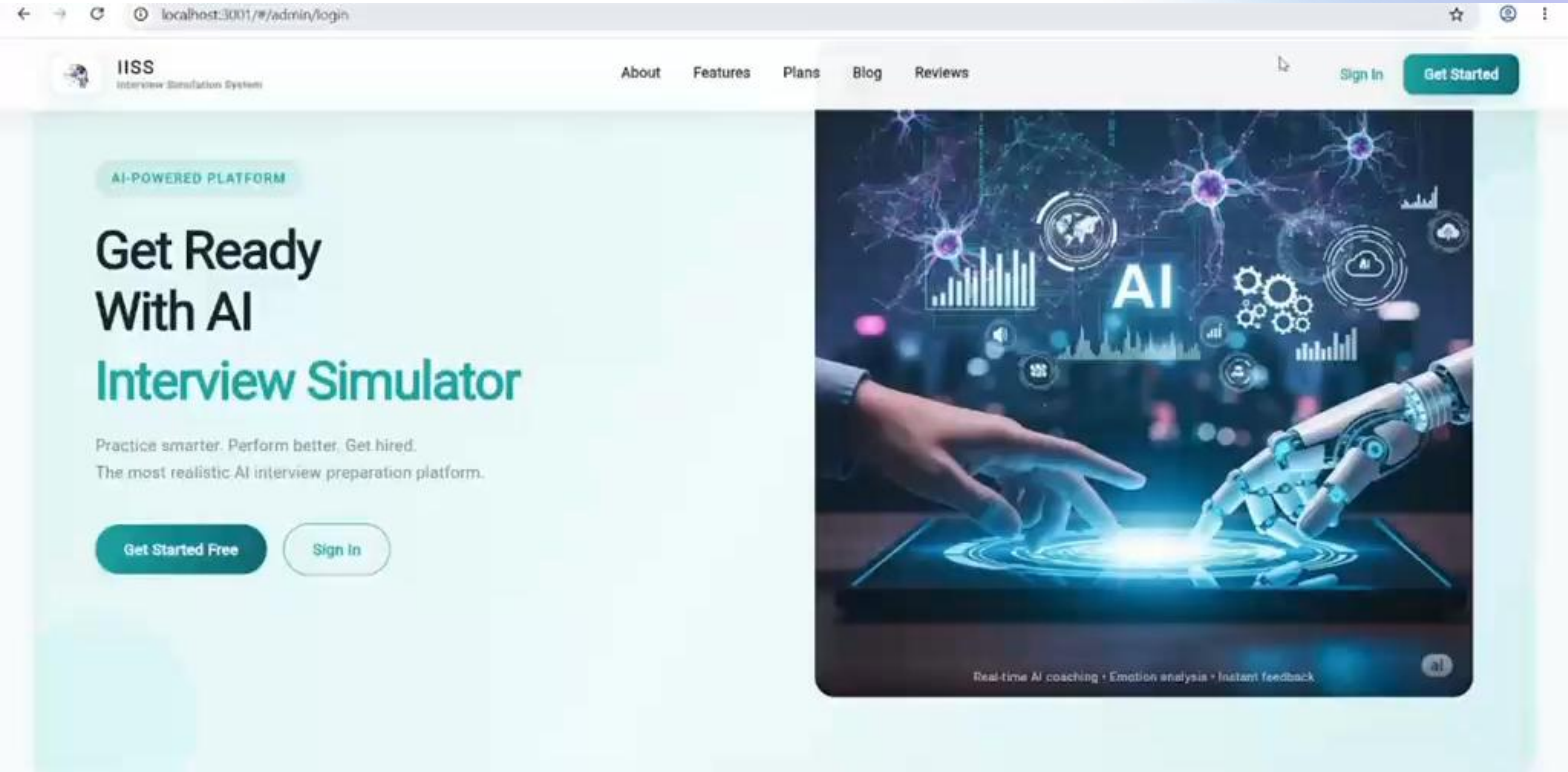
Identity Verification Scenarios

- Successful candidate identity verificatior
- Failed identity verification



System Demonstration

System Demonstration



Conclusion

Conclusion

Throughout this project, we followed a structured development journey:

• We began with the Introduction, presenting the concept of an AI-powered interview simulation platform and its role in improving interview preparation and recruitment processes.

• Through the Problem Definition, we identified the limitations of traditional interviews and existing solutions, including subjective evaluation, lack of realistic practice environments, and limited behavioral assessment capabilities.

• Based on these challenges, the Objectives chapter defined the vision of building an intelligent multimodal system capable of conducting, monitoring, and evaluating interview sessions using AI technologies.

• In Previous Work, we analyzed existing interview and recruitment systems to understand current approaches, identify research gaps, and determine the features needed to create a more comprehensive solution.

Conclusion (Cont.)

Transforming the idea into a complete system:

• During System Analysis, we translated requirements into a complete system design through use cases, ERD, OOM diagrams, system architecture, block diagrams, and workflow modeling.

• In System Implementation, we integrated all AI, security mechanisms, communication, and reporting modules into a unified platform, then conducted testing and evaluation to verify system functionality, AI performance, real-time responsiveness, reliability, and overall effectiveness.

Final Outcome ✓

A fully integrated Intelligent Interview Simulation System that combines multimodal AI analysis, real-time interview monitoring, and automated performance evaluation to support both candidates and recruiters with a realistic, secure, and intelligent interview experience.

Future Work

Future Work

Multilingual Interview Support
Support multiple languages and real-time translation for global recruitment.

Accessibility-Focused Interview Modes
Inclusive interview experiences for users with hearing or speaking limitations.

Live Text-Based Interview Responses
Allow candidates to answer through real-time text interaction.

Scalable Cloud AI Processing
Enable scalable AI workloads through efficient cloud-based processing.

Future Work

Gesture & Posture Analysis

Extend behavioral evaluation through body language and posture analysis.

Advanced Recruiter Analytics Dashboard

Compare candidates using behavioral, communication, and technical metrics.

Interview Replay & AI Summaries

Provide interview playback with AI-generated highlights and summaries.

Invisible Recruiter Monitoring Mode

Allow recruiters to monitor sessions in real time without interrupting candidates.

Future Work

Integrated Technical Assessment

Assess technical skills through interactive coding and problem-solving tasks.

Enhanced Real-Time AI Interaction

Improve live communication responsiveness and adaptive interview flow.

Assessment Environment Isolation

Restrict access to external system functionalities during assessments to enhance security, integrity, and anti-cheating enforcement on desktop app.

Thank You .